Homework 3: Support Vector Machines (SVMs)

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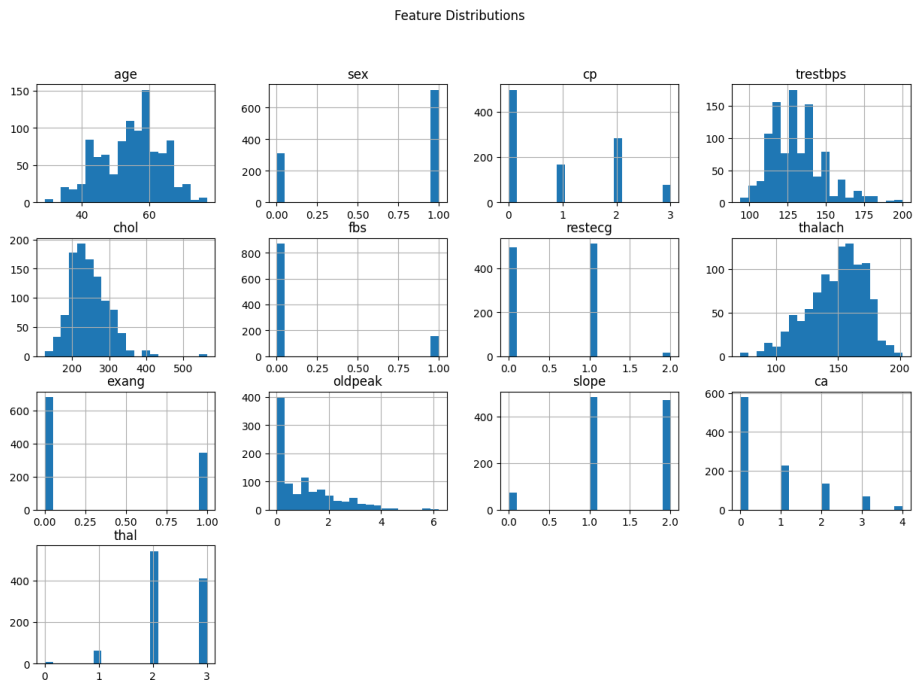
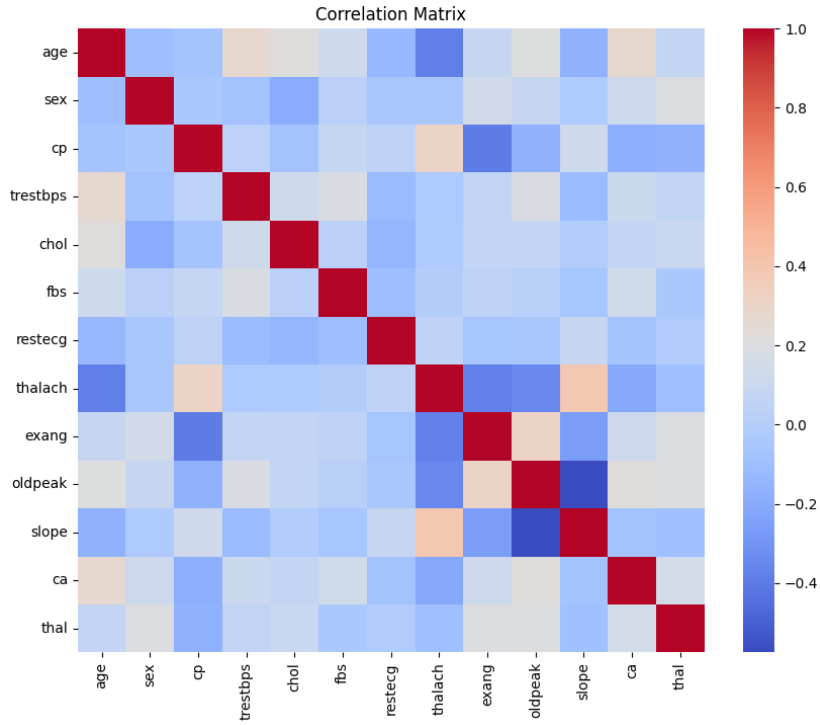
- **Data & preprocessing:** what features you used, missingness, scaling, PCA choice (if used).

HEART DISEASE DATASET

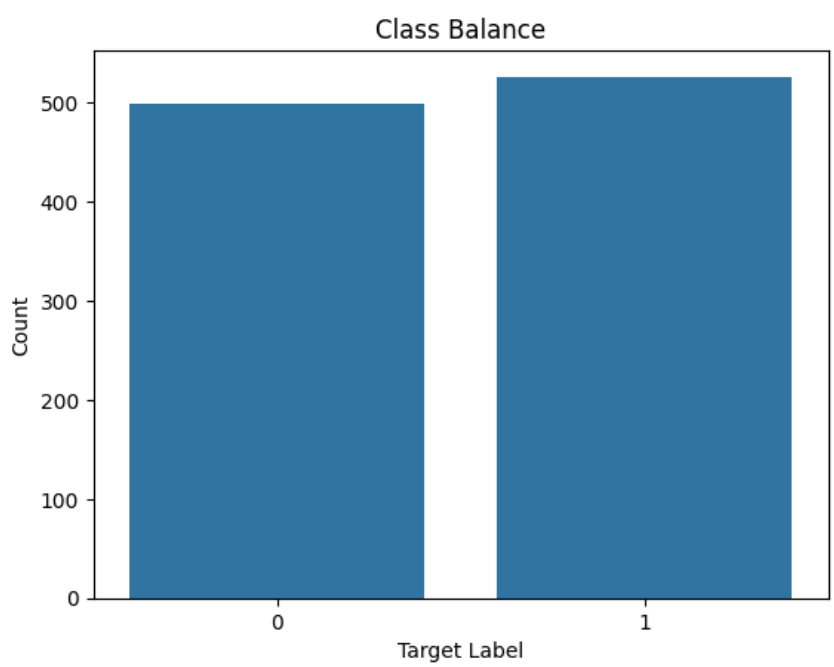
The provided heart disease dataset has 1,025 samples with 13 features. There was no missingness in any of the features so imputation was not needed. However, I did find there to be 723 duplicates, so those were removed for a total of 302 samples used for training and test data. Within the SVM classifier pipeline, I scaled the data using StandardScaler() to ensure all of the features will contribute equally.



In terms of feature balance, it seems that continuous variables follow a normal distribution with some skewness. For the binary/discrete variables, it seems some features are imbalanced. Specifically, it seems there is imbalance within the features fbs, slope, ca, thal and cp. Furthermore, it seems restecg may have an outlier at value 2.



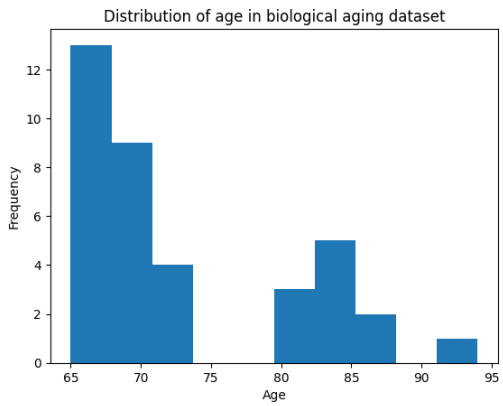
When observing the correlation matrix between all data features it appears that the highest positive correlation was observed between thalach vs. slope (corr. is approximately 0.5), whereas the highest negative correlation was observed between oldpeak vs. slop (corr. is approximately -0.5).



When examining class balance, it is evident that there is approximately the same amount for patients who have heart disease (label = 1) and those who do not have heart disease (label = 0).

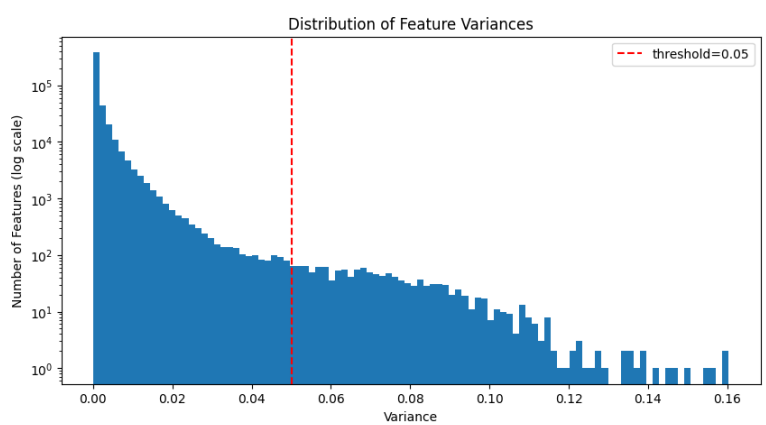
BIOLOGICAL AGING DATASET

The biological aging dataset contains 37 samples and 485,515 features, where features include sex, tissue type, and biological markers. I found no duplicates in the dataset, so no rows had to be removed. However, 265 features were found to have missing data, and missing values were filled using mean-based imputation. Within the SVM regressor pipeline, I scaled the data using StandardScaler() to ensure all of the features will contribute equally.



The distribution of age across samples is right skewed, where most participants are between 65 and 70 years old.

Given the small sample size and large number of features, I was nervous that the model may overfit the data, so I decided to subset my dataset to only look at features that tend to vary more among the samples. I set a variance threshold of 0.05 and above, and filtered the dataset to only contain 1,295 features. Note that sex and tumor type were dropped here because there was only 'male' and 'sperm' as values, and thus provide no information for the model.



- **Modeling & hyperparameters:** which kernel/params worked best for classification vs regression, and why.

HEART DISEASE DATASET

When tuning the hyperparameters for the SVM classifier, I chose C = [0.1, 1, 10, 100] for all three kernels, gamma = ["scale", "auto", "0.01", "0.001"] for rbf and polynomial, and degree = [2,3,4] for polynomial. I then got the following results, where the testing score is accuracy:

```
   kernel  cv_score  test_score  \
1     rbf  0.970732    0.985366
2    poly  0.973171    0.985366
0  linear  0.854878    0.814634

                                          best_params
1        {'model__C': 100, 'model__gamma': 'scale'}
2  {'model__C': 100, 'model__degree': 3, 'model__...
0                                  {'model__C': 0.1}

Best kernel: rbf
```

The best kernel for this dataset is rbf, with c = 100 and gamma = 'scale' as the best parameters. I think that rbf is the best choice here because heart disease classification is not linearly separable due to the complex relationship of the features in the dataset. This kernel will be able to map the data in a higher dimensional space where it can then be separable. The best parameter being the highest C indicates that the model prefers a stricter boundary, prioritizing high accuracy. For gamma, the scale parameter automatically calculates 1/(n features x the variance of X) which is balanced gamma value based on our data. This beating out the other values means that the automatic calculation helps to balance the radius of a single training sample, preventing over/underfitting.

BIOLOGICAL AGING DATASET

When tuning the hyperparameters for the SVM regressor, I chose C = [0.001, 0.01, 0.1, 1, 10, 100] for all three kernels, gamma = ["scale", "auto", "0.01", "0.001"] for rbf and polynomial, and degree = [2,3,4] for polynomial. I then got the following results, where the testing score is r^2:

```
   kernel  cv_score  test_score  \
0  linear -0.320715    0.018322
1     rbf -0.303068   -0.000308
2    poly -0.253051   -0.007246

                                          best_params
0                                {'model__C': 0.001}
1           {'model__C': 1, 'model__gamma': 'auto'}
2  {'model__C': 1, 'model__degree': 2, 'model__ga...

Best kernel: linear
```

In this case, the best kernel for the model is linear with a C parameter of 0.001. This indicates that the model is prioritizing a simple, linear relationship between age and the

gene expression values. This makes sense because we have a small sample size and large number of features, so a simple function ensures to spread and balance the weights of the different features.

- **Metrics & visuals:** include key scores + confusion matrix/ROC/PR (classification) and parity/residuals (regression).

HEART DISEASE DATASET

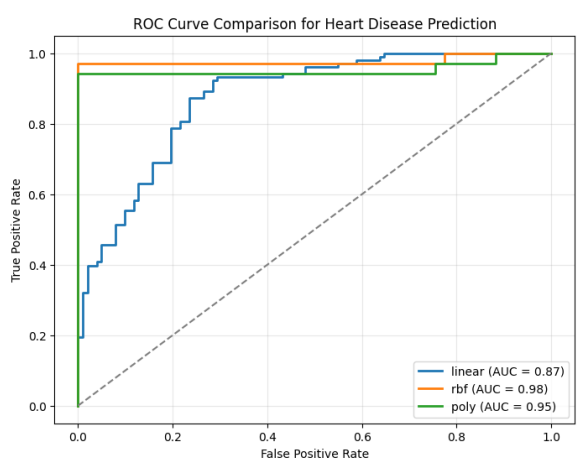
Metric Results:

	accuracy	precision	recall	f1
kernel				
linear	0.814634	0.764228	0.912621	0.831858
rbf	0.985366	1.000000	0.970874	0.985222
poly	0.970732	1.000000	0.941748	0.970000

Overall, the rbf kernel performed the best, with the highest accuracy, precision, recall, and f1. A precision equal to 1 indicates that the model was able to identify 100% of all positive cases. It is important to note that the model with the polynomial kernel also had very high metrics, which makes sense because it still captures a non-linear relationship, but cannot capture the complexity that rbf can. It makes sense that the linear kernel has lower metrics given that the heart disease classifier is not linearly separable.

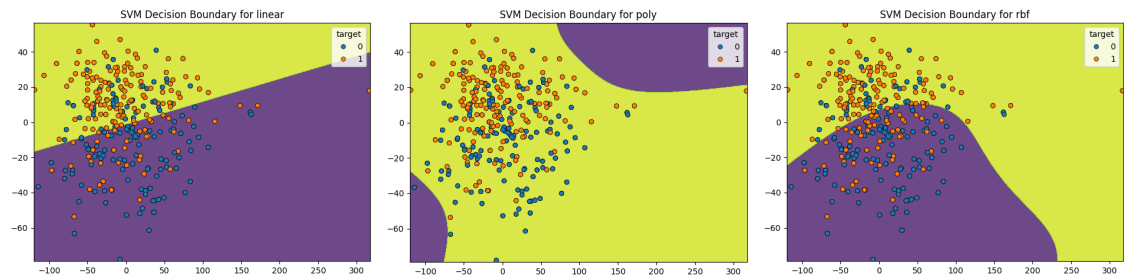
ROC Results:

```
ROC-AUC SCORES
{'linear': 0.8748334285170379, 'rbf': 0.9774414620217019, 'poly': 0.9523129640205596}
PR-AUC SCORES
{'linear': 0.871211943332198, 'rbf': 0.9873573028912834, 'poly': 0.9737471021715276}
```



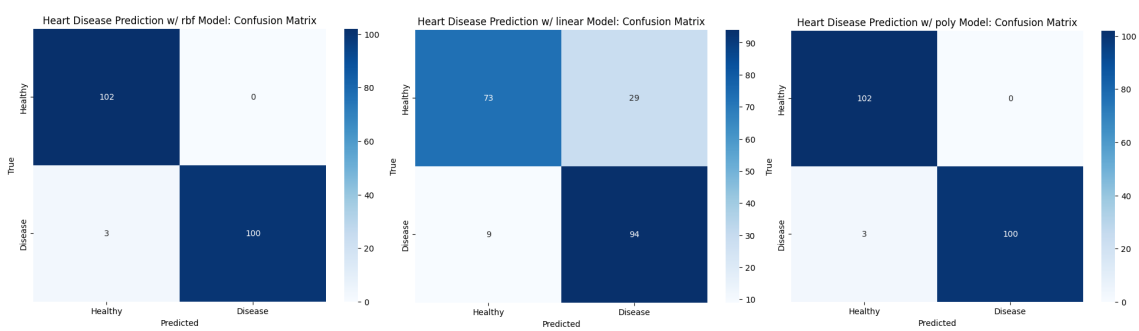
The high ROC scores for rbf and polynomial and their respective curves being in the top left corner of the graph indicates that they have high performance in their ability to distinguish those who have and do not have heart disease.

Decision boundaries:



Above are the visualized decision boundaries for the linear, polynomial, and rbf kernels. The boundary for the linear kernel struggles to separate the binary values (heart disease vs. no heart disease), with targets in both boundary areas. The decision boundary for the poly kernel seems unclear, with both target 0 and 1 in the same boundary region. Although the decision boundary for the rbf kernel does not separate all target values, it is more accurate in capturing the complex relationship in a high dimensional space.

Confusion matrices:



For the confusion matrices, both models with the rbf and poly kernels can predict all cases without disease but predicts a few patients as healthy when they truly have heart disease. For the linear model, it has a tougher time distinguishing those who are healthy from those who have disease, which makes since sense the linear model struggles to capture the complexity of all features.

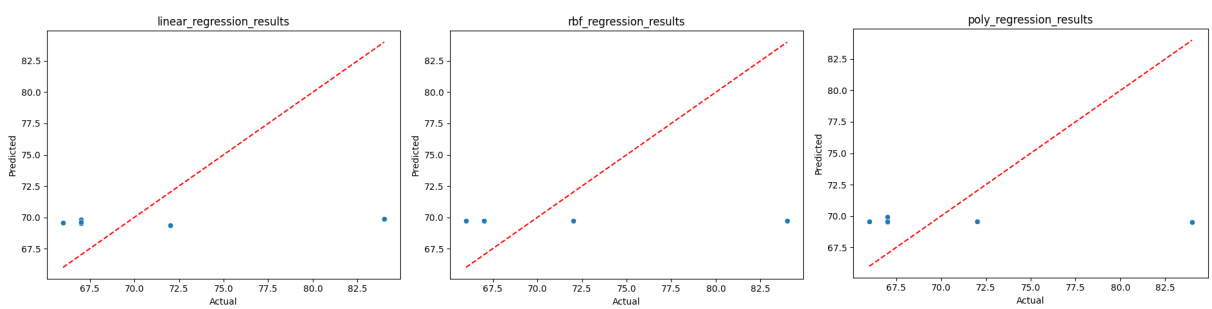
BIOLOGICAL AGING DATASET

Metric results:

kernel	mse	mae	r2
linear	5.647053	4.203031	0.018322
rbf	5.700384	4.237500	-0.000308
poly	5.720119	4.185547	-0.007246

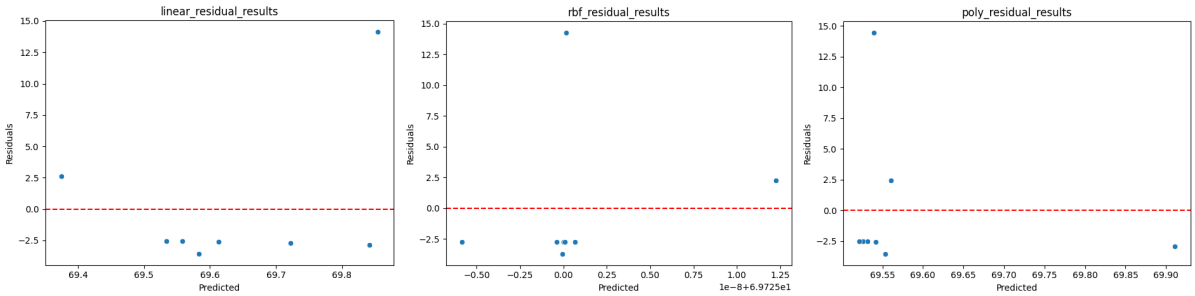
Overall, the metric results indicate that all three kernels produce a poor model fit. They all have similar MSE and MAE scores, indicating that on average all models predict age within ~4 years of the actual age. When interpreting the R-squared values, rbf and polynomial kernels have negative values, which means that the model fits the data poorly. Since the linear kernel has a positive R-squared, indicating it is the best model. Although the score is close to zero, I am not surprised by this value. We only have 37 samples to predict age with over 1,000 features, and if we have one 'bad sample', it can largely affect our model and negatively impact R-squared.

Parity plots:



When looking at the parity plot, it is evident that all three models struggle to predict the actual age. It seems they all predict that the age is approximately 69 across the board. This makes sense to me because much of the population is around 60-70. If the model is trying to give all features equal weights, and there isn't much variance across samples, the model would likely continue to guess the most frequent age.

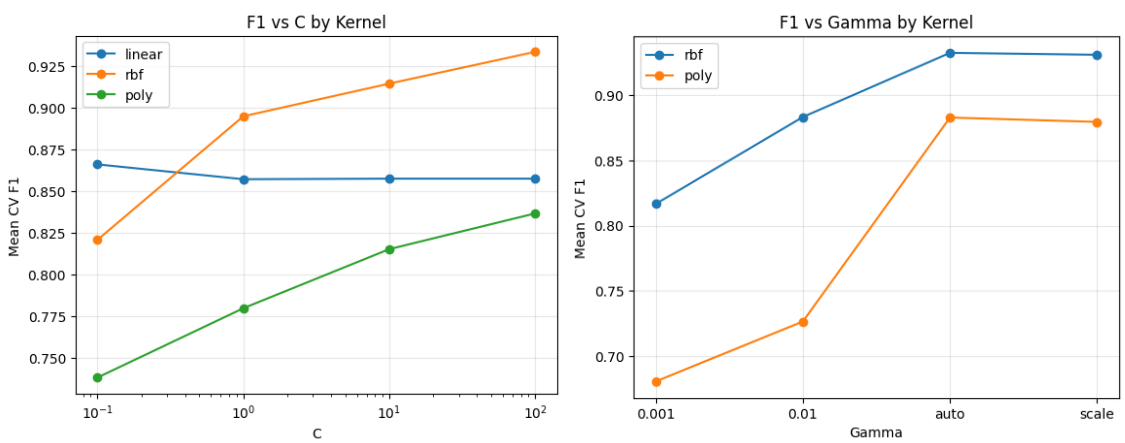
Residual plots:



The residual plot shows that all three models are predicting everyone's age as 69 years old. Again, this makes sense because a majority of the population is around this age and thus with small sample size, large feature size, and minimal variance within these features, the model is likely to default predicting the most common age seen in training.

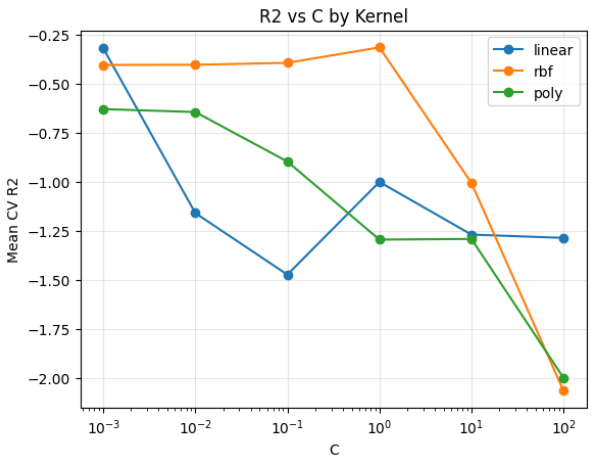
- **Sensitivity:** how results changed with scaling/PCA and with C/γ/degree.

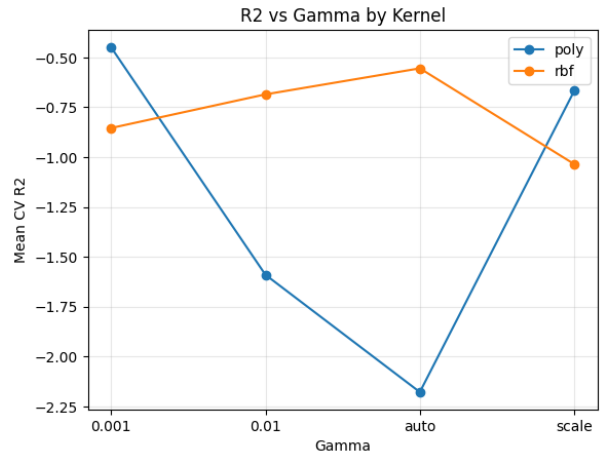
HEART DISEASE DATASET



When examining the relationship between F1 score and the C parameter, F1 score increased as the C parameter increased for the models with the rbf and poly kernels, whereas the F1 score stayed stagnant across all parameters of C for linear. This indicates that the model requires a stricter boundary for both kernels. For the relationship between F1 and gamma, both rbf and polynomial kernels perform the best when using the auto and scale gamma parameters. This indicates that the radius of each training data point must be balanced to prevent over or underfitting.

BIOLOGICAL AGING DATASET





When looking at the relationship between R-squared and C, it seems that all three kernels have different relationships. For polynomial, mean R-squared decreases as C increases because a higher C will overfit the data as the kernel tries to fit every data point. For linear, mean R-squared decreases between 0.001 and 0.1, increases again at 1, and decreases for 10 and 100, which indicates 1 may be the ideal choice to avoid being too strict or too loose with fitting the data. For rbf, R-squared remains approximately the same for 0.001 to 0.01, reaches its highest score at 1, and then drops dramatically, suggesting C = 1 balances over and under fitting the best here.

For the relationship between R-squared and gamma, both rbf and poly show different relationships. For poly, it is highest for 0.001 and 'scale' but drops at 0.01 and 'auto' as it is likely overfitting the data. For rbf, it is approximately the same for all parameters but does best with the 'auto' gamma as this setting produces a small gamma to keep the model generalized.